WANG, Suyuchen

D.O.B.: 1998/04/09 | Email: firstname.lastname at umontreal.ca
Superviser: Bang Liu (Associate Professor at Université de Montréal, Mila, CIFAR AI Chair)

EDUCATIONAL BACKGROUND

Mila / Université de Montréal, Montréal, Québec, Canada 2021/01/07-2025/12/31 (Estimated)
Département d'informatique et de recherche opérationnelle
Lab: Recherche appliquée en linguistique informatique (RALI)
Status: Ph.D. Candidate in Computer Science **GPA:** 4.18/4.3

Beihang University, Beijing, China 2016/09/01-2020/07/09
Shenyuan Honors College / National Pilot Program for Training Top Students (Everest Project)
Degree: B.Eng. (Hons.) in Computer Science and Technology **GPA:** 3.71/4.0 (Ranking in Class: 4/23)

STANDARDIZED TESTS

GRE: 326 (Verbal: 156, Quantitative: 170, Analytical Writing 4.0) Took on: 2019/08/31

TOEFL: **110** (Reading: 28, Listening: 29, Speaking: 25, Writing: 28) Took on: 2019/05/26

PUBLICATIONS

1. **Suyuchen Wang**, Jinlin Wang, Xinyu Wang, Shiqi Li, Xiangru Tang, Sirui Hong, Xiao-Wen Chang, Chenglin Wu, and Bang Liu. "Improving Context Fidelity via Native Retrieval-Augmented Reasoning". Accepted by the LCFM Workshop @ ICML 2025.
2. **Suyuchen Wang***, Jinlin Wang*, Ziwen Xia, Sirui Hong, Yun Zhu, Bang Liu, and Chenglin Wu. "FACT: Examining the Effectiveness of Iterative Context Rewriting for Multi-fact Retrieval". Accepted by NAACL 2025 Findings.
3. Bang Liu*, **Suyuchen Wang***, et al. "Advances and Challenges in Foundation Agents: From Brain-Inspired Intelligence to Evolutionary, Collaborative, and Safe Systems". arXiv:2504.01990.
4. **Suyuchen Wang***, Tianyu Zhang*, Lu Li, Ge Zhang, Perouz Taslakian, Sai Rajeswar, Jie Fu, Bang Liu, and Yoshua Bengio. "VCR: A Task for Pixel-Level Complex Reasoning in Vision Language Models via Restoring Occluded Text". Accepted by ICLR 2025.
5. **Suyuchen Wang**, Peng Lu, Ivan Kobyzev, Mehdi Rezagholizadeh, and Bang Liu. "Resonance RoPE: Improving Context Length Generalization of Large Language Models". Accepted by Findings of the Association for Computational Linguistics ACL 2024.
6. **Suyuchen Wang**, and Bang Liu. "DiffTune: A Diffusion-Based Approach to Diverse Instruction-Tuning Data Generation". Accepted by the ENLSP Workshop @ NeurIPS 2023.
7. **Suyuchen Wang**, Ruihui Zhao, Yefeng Zheng, and Bang Liu. "QEN: Applicable Taxonomy Completion via Evaluating Full Taxonomic Relations". Accepted by The Web Conference (TheWebConf, Former WWW) 2022.
8. **Suyuchen Wang**, Ruihui Zhao, Xi Chen, Yefeng Zheng, and Bang Liu. "Enquire one's parent and child before decision: Fully exploit hierarchical structure for self-supervised taxonomy expansion". Accepted by The Web Conference (TheWebConf, Former WWW) 2021.
9. Tianyu Zhang, Xinyu Wang, Zhenghan Tai, Lu Li, Jijun Chi, Jingrui Tian, Hailin He, and **Suyuchen Wang**. "STRICT: Stress-Test of Rendering Image Containing Text". Under Review.
10. Xiaoqiang Wang, **Suyuchen Wang**, Yun Zhu, and Bang Liu. "System-1.5 Reasoning: Traversal in Language and Latent Spaces with Dynamic Shortcuts". Under Review.
11. Xiaoqiang Wang, **Suyuchen Wang**, Yun Zhu, and Bang Liu. "R^3Mem: Bridging Memory Retention and Retrieval via Reversible Compression". Accepted by Findings of the Association for Computational

Linguistics ACL 2025.

12. Tianyu Zhang, Lu Li, Tongtian Zhu, **Suyuchen Wang**, Can Wang, and Yong Chen. "Metadata-Agnostic Decentralized Learning". Under Review.
13. Peng Lu, Jerry Huang, Lingfei Qian, **Suyuchen Wang**, Xueqing Peng, Jimin Huang, and Qianqian Xie. "LongFinBen: Benchmarking Long-Context Performance of Large Language Models in Financial Applications". Under Review.
14. Xueqing Peng, **Suyuchen Wang**, et al. "MultiFinBen: A Multilingual, Multimodal, and Difficulty-Aware Benchmark for Financial LLM Evaluation". Under Review.
15. Hao Xu, Xiangru Jian, Xinjian Zhao, Wei Pang, Chao Zhang, **Suyuchen Wang**, Qixin Zhang, Joao Monteiro, Bang Liu, Qiuzhuang Sun, and Tianshu Yu. "GraphOmni: A Comprehensive and Extendable Benchmark Framework for Large Language Models on Graph-theoretic Tasks". Under Review.
16. Rabiul Awal, **Suyuchen Wang**, et al. "WebMMU: A Benchmark for Multimodal Multilingual Website Understanding and Code Generation". Under Review.
17. Ahmed Masry, Juan A. Rodriguez, Tianyu Zhang, **Suyuchen Wang**, et al. "AlignVLM: Bridging Vision and Language Latent Spaces for Multimodal Understanding". Under Review.
18. Xinyu Wang, Jijun Chi, Zhenghan Tai, Tung Sum Thomas Kwok, Muzhi Li, Zhuhong Li, Hailin He, Yuchen Hua, Peng Lu, **Suyuchen Wang**, Yihong Wu, Jerry Huang, and Ling Zhou. "FinSage: A Multi-aspect RAG System for Financial Filings Question Answering". Under Review.
19. Juan A. Rodriguez, **Suyuchen Wang**, et al. "BigDocs: An Open and Permissively-Licensed Dataset for Training Multimodal Models on Document and Code Tasks". Accepted by ICLR 2025.
20. Zhiyuan Hu, Yuliang Liu, Jinman Zhao, **Suyuchen Wang**, Yan Wang, Wei Shen, Qing Gu, Anh Tuan Luu, See-Kiong Ng, Zhiwei Jiang, and Bryan Hooi. "LongRecipe: Recipe for Efficient Long Context Generalization in Large Language Models". Accepted by ACL 2025.
21. Lu Li, Tianyu Zhang, Zhiqi Bu, **Suyuchen Wang**, Huan He, Jie Fu, Yonghui Wu, Jiang Bian, Yong Chen, and Yoshua Bengio. "MAP: Low-compute Model Merging with Amortized Pareto Fronts via Quadratic Approximation". Accepted by ICLR 2025.
22. Zhiyuan Hu, Yuliang Liu, Jinman Zhao, **Suyuchen Wang**, WangYan, Wei Shen, Chao Yin, and Bryan Hooi. "CD-Pos: Long Context Generalization in LLMs Through Continuous and Discrete Position Synthesis". Accepted by the LCFM Workshop @ ICML 2024.
23. Hossein Rajabzadeh, **Suyuchen Wang**, Hyock Ju Kwon, and Bang Liu. "Multimodal Multi-Hop Question Answering Through a Conversation Between Tools and Efficiently Finetuned Large Language Models". Accepted by the ENLSP Workshop @ NeurIPS 2023.
24. Peng Lu, **Suyuchen Wang**, Mehdi Rezagholizadeh, Bang Liu, and Ivan Kobyzev. "Efficient Classification of Long Documents via State-Space Models". Accepted by EMNLP 2023.
25. Yaowei Zheng, Richong Zhang, **Suyuchen Wang**, Samual Mensah, and Yongyi Mao. "Anchored model transfer and soft instance transfer for cross-task cross-domain learning: a study through aspect-level sentiment classification". Accepted by The Web Conference (TheWebConf, Former WWW) 2020.

PATENTS

1. "A multi-round dialogue system for flight check-in scenarios". CN111858888B, Issued May 30, 2023.
2. "A data processing method, device and storage medium". CN113568901B, Issued Apr 19, 2024.
3. "Information matching method, device, equipment and storage medium". CN113821610A, Filed Dec 21, 2021.
4. "Node addition method, device, equipment and storage medium based on taxonomy". CN114281919A,

Filed Apr 5, 2022.

5. "Taxonomy expansion method, model training method, device, equipment and storage medium". CN114282066A, Filed Apr 5, 2022.

TEACHING EXPERIENCES

Teaching Assistant for **IFT 6289 - Natural Language Processing with Deep Learning** Winter 2024
Teaching Assistant for **IFT 6759 - Advanced Machine Learning Projects** Winter 2023
Teaching Assistant for **IFT 6289 - Natural Language Processing with Deep Learning** Winter 2022

INTERNSHIP EXPERIENCES

ServiceNow Research, Montreal, Quebec, Canada 2025/01-Now
Visiting Researcher
Working on vision-language model research; Published 1 paper as co-first-author in ICLR 2025, and contributed as core member to 1 ICLR 2025 paper and 1 ICML 2025 submission

Huawei Noah's Ark Lab, Montreal, Quebec, Canada 2022/06-2024/03
Natural Language Processing (NLP) Research Intern
Published 3 papers with colleagues in ACL and the ENLSP workshop @ NeurIPS 2023 about Long Context Modeling and Tool-assisted LLMs

Tencent Jarvis Lab, Shenzhen, Guangdong, China 2020/07-2021/07
Natural Language Processing (NLP) Research Intern
Published 2 papers in The Web Conference about Taxonomy Expansion and Taxonomy Completion

HONORS & AWARDS

1. Bourses d'études dans le domaine de l'intelligence artificielle de l'Université de Montréal (10,000 CAD) (2021)
2. The Honor of Outstanding Undergraduate of Beihang University (2020)
3. The Honor of Merit Student of Beihang University (2019, 2018 & 2017)
4. The Grand Prize Scholarship for Outstanding Social Work Performance of Beihang Uni. (2019)
5. The First Prize Scholarship for Outstanding Academic Performance of Beihang Uni. (2019)
6. The First Prize Scholarship Outstanding Competition Performance of Beihang Uni. (2019)
7. The Meritorious Winner of the 2018 Mathematical Contest in Modeling (MCM)
8. The Second Prize of the 2018 Beijing Contemporary Undergraduate Mathematical Contest in Modeling
9. The First Prize of the 2018 Mathematical Modeling Contest of Beihang University
10. The Grand Prize Scholarship for Outstanding Competition Performance of Beihang Uni. (2018)
11. The First Prize Scholarship for Outstanding Social Work Performance of Beihang Uni. (2018)
12. The First Prize Scholarship for Outstanding Social Practice Performance of Beihang Uni. (2018)
13. The Third Prize of Band C group in 2017 National English Competition for College Students
14. The Third Prize in the 2017 Mathematical Modeling Contest of Beihang University
15. The Third Prize Scholarship for Outstanding Academic Performance of Beihang Univ. (2017)
16. The Third Prize Scholarship for Outstanding Literature and Art Performance of Beihang Uni. (2017)
17. The Scholarship for Excellent Freshman of Beihang University for the 2016 Academic Year

ACADEMIC VOLUNTEER WORKS

Workshop Organizer - Multimodal Knowledge and Language Modeling (MKLM) @ IJCAI 2025 2025/08
Workshop Organizer - New in ML Workshop @ ICML 2025 2025/07
Mila Recruitment Evaluation Committee 2024/11
Mila Recruitment Evaluation Committee 2023/11
Student Volunteer - the 30th International Joint Conference on Artificial Intelligence (IJCAI 2021) 2021/08

OTHER SKILLS

Computer Skills: Proficient in programming (familiar with C, C++, Java & Python); Received the Certified Software Professional Certificate from the China Computer Federation (CCF-CSP) for C/C++ (top 9.1%)